

Void Banshee Exploitation of CVE-2024-38112

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Introduction

Amidst the constant innovation and progress in technology, there exists a useful tool known as threat intelligence. Threat intelligence can go by several different names such as threat intel or cyber threat intelligence. Threat intelligence is an important field within cybersecurity.

According to the Blue Goat Cyber website, threat intelligence “refers to the collection, analysis, and interpretation of information about potential cyber threats, their capabilities, and intentions.” Such a field of knowledge has become indispensable to stakeholders looking to protect their assets from threat actors.

Threat intelligence provides information on vulnerabilities and exposures. In response to the detection of these vulnerabilities and exposures, patches and updates are expedited to eliminate them. This process is then recorded into an international framework that records all known cybersecurity threats. Better known as a CVE or Common Vulnerabilities and Exposures. These CVE’s are given unique ID’s which are then given a score based on the Common Vulnerability Scoring System, which is used to evaluate its threat level. The recording of these CVE’s is very important and requires coordination between researchers and manufacturers. CVE’s ensure that information on vulnerabilities and exploitations is dispersed and distributed in an orderly manner. Throughout this report, we shall look into a recent CVE and discuss its impact

CVE Details

CVE-2024-38112 was published on July 9th, 2024 and originated from the CNA: Microsoft Corporation. The title of our chosen CVE is listed as “Windows MSHTML Platform Spoofing Vulnerability”. Due to lack of any information provided on the CVE website, we look

towards the CWE for more information on the vulnerability. The common weakness enumeration (CWE) present in the report is CWE-451: User Interface (UI) Misrepresentation of Critical Information. In which the UI does not accurately and properly portray critical information to the user. This allows for such information to be spoofed or hidden. CWE-451 is a component of phishing attacks that trick the User into providing sensitive information to a “trusted” source. The affected system may find its UI displaying incorrect indicators, icon manipulation, or visual truncation among other misrepresentations (CWE).

CVE-2024-38112 was given a score of 7.5 under CVSS and placed as HIGH severity. The following products of Microsoft were affected by the CVE: Windows 10 Versions 21H2, 22H2, 1507, 1607, Windows 11 Versions 22H2, 23H2, Windows Server 2016, Windows Server 2016 (Server Core Installation), and Windows Server 2008 Service Pack 2. Between all these products a mixture of the 32-bit Systems, ARM64-based Systems, and x64-based Systems were among the affected platforms. This extensive list of products that were impacted by CVE-2024-38112 only serves to further highlight its widespread reach.

Deciphering the vector string recorded provides us with more knowledge on our CVE.

CVSS:3.1/AV:N/AC:H/PR:N/UI:R/S:U/C:H/I:H/A:H/E:F/RL:O/RC:C

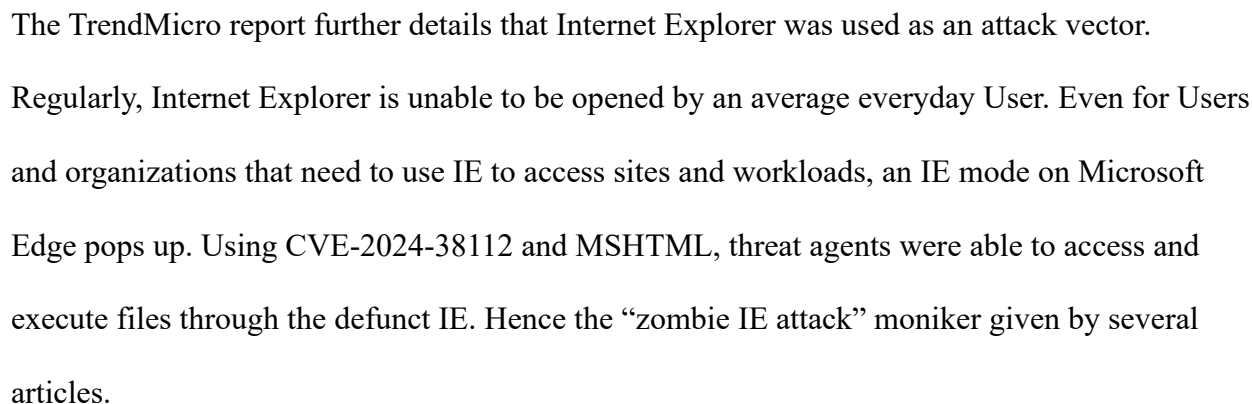
From this string we can ascertain that the attack vector is categorized as Network. This means the vulnerability was exploited through Network access also known as being remotely exploitable. The attack complexity being high implies that it requires considerable effort on behalf of the attacker to carry out a successful exploit. No privileges are required; therefore, the would-be attacker can be unauthorized and does not require access to any settings. This vulnerability also requires User interaction, as gleaned by the description of CWE-451. The rest Base Score

Metrics in the string refer to the Scope: Unchanged, Confidentiality Impact: High, Integrity Impact: High, and Availability Impact: High. Furthermore, the Temporal Score Metrics are also listed: the Exploit Code Maturity: Functional Exploit exists, the Remediation Level: Official Fix, and Report Confidence: Confirmed. The Common Vulnerability Scoring System Calculator proved handy in interpreting the vector string.

Event Analysis

CVE-2024-38112 was not the first CVE I had chosen; however, it proved difficult to identify corresponding cybersecurity events. Therefore, I chose to work backwards and stumbled upon the exploits of the Advanced Persistent Threat (ATP) group, Void Banshee. According to the Hacker News, Void Banshee made use of “a security flaw in the Microsoft MHTML browser engine as a zero-day to deliver an information stealer called Atlantida.” Another headline coined it a “zombie IE attack” (CSO).

The articles described how Users were tricked into downloading malicious files which were disguised as PDF versions of reference books. These files were circulated through Discord servers, online libraries, cloud-sharing websites, and other sectors (DarkReading). The malware, Atlantida, is a stealer which extracts sensitive data like passwords and cookie information. Furthermore, the malware can capture the User’s screen facilitating procurement of comprehensive system information. Once the malicious files were downloaded and the User attempted to open them, an attacker-controlled website was opened through Internet Explorer (IE). It is important to note that even though IE has been disabled in later versions of Windows 10 and throughout versions of Windows 11, remnants of Internet Explorer remain in the system.



Void Banshee is reported to have targeted entities throughout North American, European and Southeast Asian countries. Their main goals in their exploits are information theft and financial gain. Given all this information, it can be concluded that patch management is critical to avoid falling victim to vulnerabilities such as CVE-2024-38112. Especially when there is the

technical risk of data loss and financial risk of monetary deficit. In addition to patch management, it is important to brush up on recognizing spoofing and phishing scams. This point should be particularly significant in the corporate world, where human risk is rife, and cybersecurity training should be mandatory.

Correlation

“CVE-2024-38112 was used as part of an attack chain by the advanced persistent threat (APT) group Void Banshee (TrendMicro).” It was a critical enabler in their exploits and had been used in the wild as ZDI-CAN-24433. It was a zero-day bug that had been in use for at least 18 months prior to Microsoft’s patch release (DarkReading). A zero-day vulnerability is a security flaw in a system or software that is actively being used in exploits before any mitigation exists. Unfortunately, this vulnerability is one of many zero-day exploitations that plagued 2024. In a threat intelligence analysis, Google reports to have tracked “75 zero-day vulnerabilities exploited in the wild in 2024.” 44% of which targeted enterprise products. To be more specific, 20 of these targeted security and network products like Ivanti, Palo Alto Networks, and Cisco. These figures highlight the need for proactive vulnerability management and mitigation.

Conclusion

The zero-day exploit of CVE-2024-38112 by Void Banshee, reveals how MSHMTL flaws can bypass security controls. It emphasizes the necessity of rapid patch adoption and enhanced phishing defenses among other cybersecurity strategies and protocol. Most notably under the threat of data theft and financial harm. Just as there is much good that can arise from the

progression of technology, evil too can be fomented. In a society so reliant on the internet, it is important to stay up to date on emerging technologies and exploitations.

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